

UTM teacher innovates to save soldiers' lives

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Photo :- Anup Gurung at Guest House of DRDO, Naval Systems and Materials at Vishakhapatnam.

Anup Gurung, a faculty in the Department of Chemistry, **University of Technology and Management, Shillong** has developed a jacket which can be used in any climatic conditions -- hot or cold. He has named the product **HX-1 (Heat eXchange Jacket)**. The jacket uses the simple principle of heat thermodynamics on a material that is easily available.

Anup who made a presentation before a DRDO (Defence Research & Development Organization) team headed by Director General of Naval Systems and Materials at Vishakhapatnam in the first week of March, has been invited to write a detailed proposal. Anup, is confident that his innovation can help a large number of Indian soldiers posted in extremely inhospitable regions like Siachen Glaciar and Thar desert of Rajasthan. In Siachen, the highest battlefield in the world, where the temperature dips far below freezing point – up to -40°C and more – the casualty rate has been very high because of frost bite rather than enemy bullets.

Anup, who hails from Darjeeling, has been working on the jacket for nearly three years, making it an all-weather cost effective product. His creation caught the eye when it won the second prize in Bee Fest 2012 held at Tashi Namgyal Academy in Sikkim from among participants from 183 institutions.

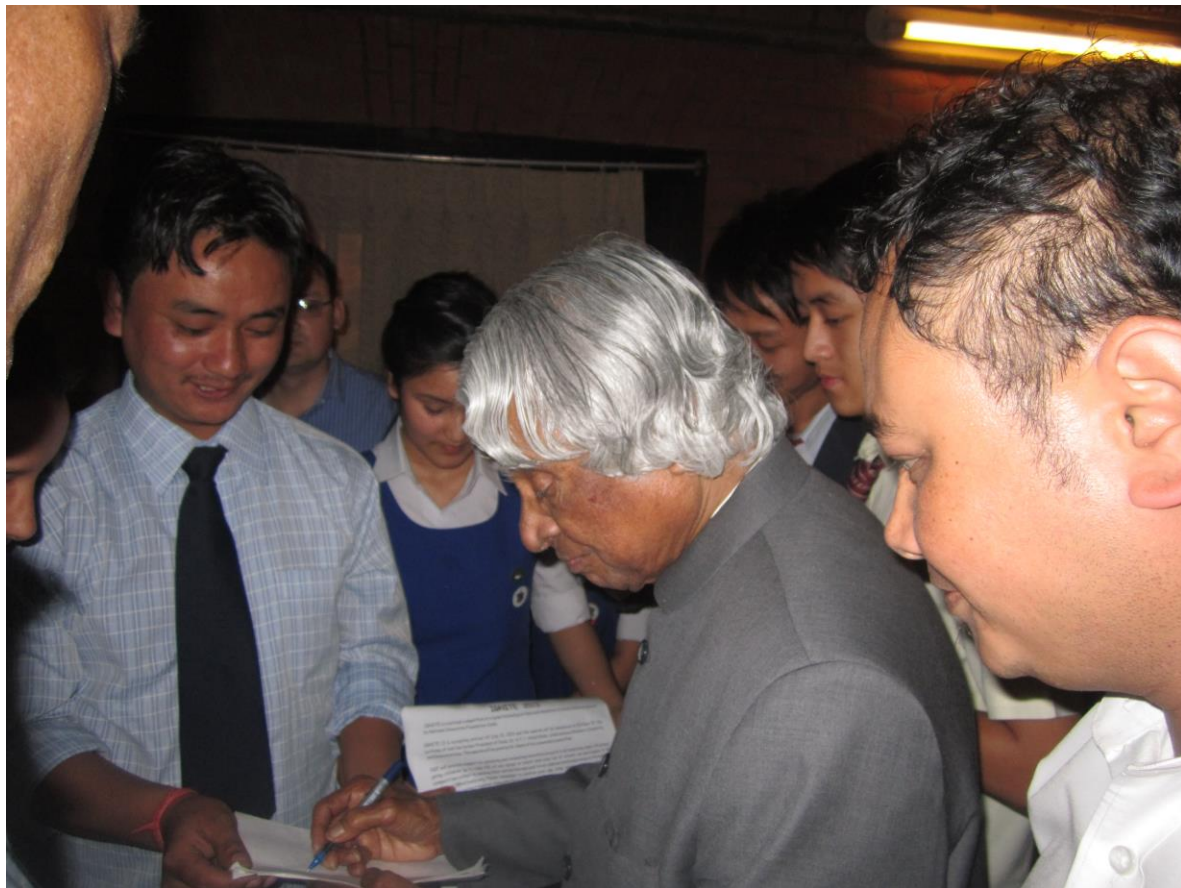


Photo : Anup Gurung with former Indian President APJ Kalam Azad at National Innovative Foundation Award Ceremony at IIM- Ahmedabad

The jacket can maintain the body temperature at any desired value ranging from - 10° C to 55°C or more. For example body temperature can be kept at 0°C, 20°C or 45°C thus making one feel warm in coldest regions of the country like North Sikkim and upper reaches of Uttarakhand, Himachal, J&K and Arunachal Pradesh or feel cold in the very hot areas like Rajasthan and other areas of extreme heat.

Anup reveals that a person suffering from *hypothermia* (low heat content in the body) or *hyperthermia* (high heat content in the body) can be effectively taken care with HX-1. The young scientist is sure that with little more work, his invention can be commercialized. The product produces no waste and is totally environmentally friendly, he claims.

Anup who is pursuing his Doctorate is all set to submit his thesis on Chemical Sensor and Anti-cancer Drugs. Coming from a humble background, Anup believes that quality education, like the rays of sun, should be available to everybody irrespective of social stature.

There will be three types of such clothing's HX-1, HX-2 and HX-3 for the time being which will be used in different environment by the Indian soldiers in particular.



Photo :- Anup wearing HX-1 at his apartment.



Worked with the Hindustan Times for 19 years, Special Correspondent from 90-97. Covered many national and international assignments, prominent being 1984 Los Angeles Olympics, World Cup Hockey in London 1985, World Cup Athletics in Helsinki, Operation Blue Star and Black Thunder, IPKF operations in Sri Lanka.

Also reported for BBC, Reuters and Voice of America. Did a cover story for TIME magazine.

Worked as the COO of a 24-hour TV news channel.

Have been instrumental in setting up and heading five leading Media Departments/Institutes of the country

Was Proctor Amity University and Director Amity School of Communication.

Also headed one of the most modern Media and Communication Institute in the country -- MassCoMedia

Specialties

Academics and administration, corporate communication (CSR specialist), strategic marketing, training and development, experiential learning, technical and creative writing, linkages, tie-ups and facilitations, brand development, career counselling, competitive Intelligence and corporate warfare. Teaching: Corporate PR, media marketing, TV production techniques, print media, content development

Note :- The report has not been published anywhere but distributed among the university members and students.